FEVEN ABEBE BEJIGA

Applied AI/ML Engineer | Health Informatics | Clinical Decision Support | Responsible AI
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PROFESSIONAL SUMMARY

Applied AI/ML practitioner with a BSc in Health Informatics (CGPA 3.86/4.00), an MSc in Artificial Intelligence in progress at Addis Ababa University, and 3+ years of digital health experience in EMR, DHIS2, and clinical decision-support workflows, including training 100+ healthcare staff. Builds AI decision-support prototypes using machine learning, computer vision, retrieval-augmented LLMs, and time-series forecasting, with emphasis on data quality, evaluation, explainability, and human oversight. Built a demand-forecasting model 19% better than a seasonal-naive baseline in a hackathon (synthetic data). First author of a federated learning paper slated for PanAfriCon AI 2026.

TECHNICAL SKILLS

- **Programming and Data:** Python, SQL, Java, C++, JavaScript, pandas, NumPy, SciPy, Matplotlib, Seaborn
- **Machine Learning and AI:** PyTorch, TensorFlow/Keras, scikit-learn, CatBoost, LightGBM, computer vision, NLP, classification, regression, time-series forecasting, feature engineering
- **AI and LLM Integration:** Gemini API, large language models, retrieval-augmented generation (RAG), FAISS, Sentence Transformers, semantic search, prompt engineering
- **Evaluation and Responsible AI:** ROC-AUC, AUC-PR, precision, recall, F1-score, cross-validation, model comparison, SHAP, LIME, preliminary fairness assessment, prediction intervals
- **Backend and Integration:** Python, FastAPI, REST APIs, Streamlit, Docker, Git, GitHub Actions, automated testing, continuous integration
- **Open-source Health Systems:** Bahmni, DHIS2, MySQL, EMR
- **Digital Health:** WHO Digital Adaptation Kits, clinical decision support systems (CDSS), data validation, indicator reporting, data integration
- **Documentation:** Technical documentation, user guidelines, system user training, knowledge transfer

PROFESSIONAL EXPERIENCE

Big Data / Data Science Fellow | Ethiopian Statistics Service (ESS) | October 2026 – Present

Current placement; activities are exploration and review, not completed implementations.

- Mapping the structures, indicators, and data-quality requirements of national statistical data systems and survey datasets, including agricultural (AgSS) and demographic and health (EDHS) surveys.
- Assessing 4 integration methods (data reconciliation, data fusion, spatial joins, seasonal adjustment) and harmonization considerations across 3 statistical domains: agricultural, demographic, and health.
- Identifying candidate data science and analytics applications to support evidence-based planning and decision-making.

Digital Health Developer | CDHi, University of Gondar | July 2023 – January 2026

- Implemented and maintained open-source health information systems, including Bahmni and DHIS2, across 3 healthcare facilities, covering EMR reporting and routine health data workflows.
- Co-implemented WHO Digital Adaptation Kit (DAK) clinical workflows in the EMR at 3 facilities (University of Gondar Comprehensive Specialized Hospital, Han Health Center, Shimbit Health Center), configuring decision logic, validation rules, mandatory fields, skip logic, and automated calculations.
- Authored an EMR reporting module in SQL, used at 3 facilities, generating aggregated indicators for 8+ service areas (ANC, FP, NCD, ambulance service, inpatient, outpatient, PICT, TB); administered MySQL databases, safeguarding data integrity and resolving issues.
- Configured and supported DHIS2 dashboard-based reporting across 3 workstreams: Tracker workflows for the Health and Demographic Surveillance System (HDSS), aggregate reporting for the Integrated Supportive Health Office (ISHO), and facility assessment data from Excel into DHIS2, with validation and program rules set up across many data elements.
- Trained 100+ nurses, physicians, and other staff, with workflow guidance, troubleshooting, and user documentation.

Digital Health Intern / Fellow | CDHi, University of Gondar | November 2022 – June 2023

- Tested and documented digital health system implementations; provided user support, health data management, system configuration, data validation, and troubleshooting alongside healthcare professionals and technical teams.

SELECTED AI/ML PROJECTS

Work from 2024 to 2026 spanning MSc coursework, independent engineering, hackathon participation, and research.

AgriMind — AI Decision Support for Crop Disease (early-stage prototype)

Python · TensorFlow/Keras · EfficientNet · Sentence Transformers · FAISS · Gemini API · Streamlit

- Developed an AI-powered prototype combining maize and wheat disease classification (EfficientNet), FAISS-based retrieval, and the Gemini API to deliver evidence-grounded information to agricultural extension workers.
- Curated a knowledge base of 29 records from 8 sources, organized by year, region, evidence type, and confidence, with semantic retrieval (sentence embeddings, FAISS) prioritizing evidence for the predicted disease and crop.

CoffeeGuard AI — Coffee Leaf Disease Detection (training project)

Python · PyTorch · EfficientNetV2-B0 · ONNX Runtime · FastAPI · Streamlit · Docker

- Contributed to training an EfficientNetV2-B0 transfer-learning model for four coffee-leaf classes (Healthy, Cercospora, Leaf Rust, Phoma) and helped oversee data cleaning.
- Team pipeline cut a 12,000-file public dataset to 2,520 unique photos (removed 902 exact duplicates, 69 label conflicts, and author-made augmentations), grouped near-duplicates by hash and image similarity, and split group-aware so no leaf crosses train, validation, and test.
- Team reduced photo-setup bias with background-swap and quality-equalisation augmentation.
- Team model handles uncertainty with temperature scaling (ECE 0.097 to 0.014), conformal prediction, and a KNN out-of-distribution gate that rejected 180 of 182 non-coffee test images; macro-F1 0.974 on 379 held-out photos. Accuracy fell to 63% on another dataset; not yet validated on farm photos.

DHIS2 Metadata Intelligence (prototype)

Python · FastAPI · React · TypeScript · SQLAlchemy · Sentence Transformers · scikit-learn · Docker

- Created a read-only prototype that flags potentially duplicate DHIS2 metadata and shows the downstream effects of changing or deleting it, combining 4 similarity signals (text, embeddings, metadata compatibility, structure) with dependency analysis.
- Designed a human-review workflow separating potential matches from confirmed decisions, and added an evaluation framework to compare alternative similarity approaches.

Ride Demand Forecasting — Qiyas Data Science & AI Hackathon (team of 6)

Python · pandas · LightGBM · scikit-learn · ARIMA · FastAPI · Next.js

- Co-developed hourly demand forecasting across 12 city zones (10 months of synthetic ride-hailing data plus weather and events); corrected 28 data-quality issues, including a UTC-to-local-time mismatch that raised the rain–demand correlation from 0.02 to 0.27.
- Compared 10 forecasting approaches with 5 rolling-origin folds and feature-ablation experiments; a tuned LightGBM model (Poisson objective) reached an RMSE of 8.09 on a held-out fortnight, 19% better than the seasonal-naive baseline. Synthetic data; results are experimental, not operational impact.

House Price Prediction System

Python · scikit-learn · CatBoost · SHAP · FastAPI · Streamlit · Docker · GitHub Actions · live demo

- Assembled an end-to-end pipeline on the Ames housing dataset (1,460 homes): benchmarked 12 regression algorithms and tuned CatBoost with 5-fold grid search, reaching R^2 0.937 (RMSE about $18.6K) on a 292-home held-out test set.
- Added SHAP explanations, a FastAPI API, and a live Streamlit app; containerized with Docker, with pytest tests and GitHub Actions CI that builds and publishes both images.

RESEARCH AND CONFERENCE

Lightweight Metaheuristic Optimization for Client Selection in Resource-Constrained Federated Learning Systems

Bejiga, F. A., Yassin, Z., and Gizachew, B. (Addis Ababa University) · First author; accepted for presentation at PanAfriCon AI 2026

- Implemented and evaluated 3 metaheuristics (Genetic Algorithm, Particle Swarm Optimization, Simulated Annealing) for client selection against a random baseline in a PyTorch federated-learning setting (FedProx, non-IID data).
- Tested on 3 datasets (FEMNIST, CIFAR-10, CIFAR-100) with 5 random seeds and 100 communication rounds.

EDUCATION

MSc in Artificial Intelligence — Addis Ababa University (AAiT) | November 2024 – Present

Proposed thesis: *Selective Neuro-Symbolic Prediction with Proof-Based Acceptance under Concept and Rule Shift*

BSc in Health Informatics — University of Gondar | CGPA 3.86/4.00 | 2022

PROFESSIONAL TRAINING

- **Data Science and AI Engineering** — Qiyas, Addis Ababa University (World Bank-supported) | May – September 2026
- **SAFEE KAIM** — Kifiya AI Mastery Training Program, 10 Academy Women-Only Cohort (Mastercard Foundation-supported) | June – August 2026
- **GCI World** — Data Science, Matsuo-Iwasawa Laboratory, University of Tokyo | June – August 2026

LANGUAGES

Amharic — Native | English — Fluent